

Hudson Ferraz Oliveira

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FULL-STACK DEVELOPER

ABOUT ME

Full-stack developer specializing in browser-based robotics simulation and high-frequency trading systems. I ship production client systems under hard platform constraints (WebAssembly physics, neural-network inference, mobile browsers) and exchange integrations for long-running trading bots. Strong in TypeScript and Python, with focus on reliable automation, clear observability, and maintainable code.

EDUCATION AND PERSONAL DEVELOPMENT

- Information Systems | UFVJM – Diamantina, MG | 2014 – (incomplete)
- English Fluent | Official EF SET Certificate (C2 Proficient)
- Portfolio | hudson-portfolio-website.vercel.app

PROFESSIONAL EXPERIENCE

Full-Stack Developer | Runibi (runibiLABS)

Jul 2026 - Sep 2026

Built a browser-based robotics simulation platform where users train humanoid robots through game missions. Real-time MuJoCo physics (WebAssembly) and ONNX neural-network control policies run entirely on the client, on desktop and mobile.

- SimAs MuJoCo Game:** React/TypeScript game with Three.js/React Three Fiber driving a Unitree G1 humanoid via mujoco-js physics and ONNX Runtime Web policy inference. Built a 3D Worldmap for mission selection, NPC dialogue with cinematic camera cuts, a finale sequence, and mobile touch controls (continuous D-pad with snap assist).
- Mobile & iOS/WebKit Hardening:** Made WASM physics + neural inference reliable on iPhone/Android. Fixed OOM crashes between missions, set up COOP/COEP for SharedArrayBuffer in iframes, handled WebKit threading limits and Cloudflare WASM cache poisoning, and improved long sessions (lazy code-splitting, lighter materials, tighter physics catch-up, pause when hidden).
- Crash Diagnostics & Platform:** Client crash telemetry with breadcrumbs, write-ahead death reports that survive tab kills, and a Supabase crash pipeline plus opt-in ?debug=1 console. Set up the Bun monorepo, FastAPI backend, Supabase auth, and Cloudflare Pages deploys.

Impact: robot model download ~58% smaller (74→31 MB, lossless OBJ→STL); 200 Hz physics + 50 Hz NN control in-browser; 50% lower worst-case physics catch-up (100→50 ms); peak mobile asset downloads 8→3 (OOM fix); GamePage JS -60% / app code -20% gzip via lazy loading; ~1,450 tests kept green; 6 missions + finale.

Full-Stack Developer | Funttastic Company

Jan 2024 - Jun 2026

Contributed to high-frequency trading and market-making platforms, integrating exchanges, liquidity algorithms, observability, and distributed infrastructure across multiple blockchains.

- Hummingbot Open-Source Contribution:** Built custom CLOB/AMM exchange connectors, improving multi-exchange communication and reliability for distributed trading systems.
- Rujira Blockchain HFT Bot:** Architected a full HFT bot for Rujira Trade (FIN) with adaptive spread/skew engines, GraphQL + Cosmos SDK integration, secure wallet management, and real-time P&L monitoring.
- “Barracuda” HFT Platform:** TypeScript/Bun market-making system with Raydium/Rujira/CCXT adapters, CLMM safety logic, structured logging, Telegram alerts, and long-running dedicated-server validation.